

IMPACT OF WEATHER AND TRAFFIC ON AIR POLLUTION: AN UNCERTAINTY AND SENSITIVITY ANALYSIS

By Diep Truong | System Dynamics with Applications | February 2026

AI declaration: This model was built with support of ChatGPT as a learning tool. ChatGPT was used to help understand concepts and to assist in generating and fixing code.

Model Description

The purpose of this model is to examine the impact of weather on traffic and, ultimately, how weather and traffic affect air quality, and it was built on Jupyter Notebook. This model can be used to assist decision-makers in traffic and air pollution policy management, or by those concerned with the environment. For example, how can we use transportation efficiently without polluting the air? The model identifies the most important factors affecting the concentrations of Nitrogen Dioxide (NO_2) and Black Carbon (BC). The data used to build this model covers nearly three years in hourly time, combined with:

- Traffic volume data from traffic counter ID 32 (Helsinki Region Infoshare, 2026).
- Weather data from Kaisaniemi (Finnish Meteorological Institute, 2026).
- Air pollution data from Kallio (Helsinki Region Environmental Services, 2026).

Table 1 below shows inputs, outputs, and decision variables:

Decision variables	Traffic volume reduction level, implemented by scaling the traffic volume
Uncertain variables	Traffic volume (value), wind_speed, precipitation, snow_depth, temperature, pressure, humidity, cloud_cover, hour, dayofweek, month
Output variables	Nitrogen dioxide (NO_2) concentration [$\mu g/m^3$], Black carbon concentration [$\mu g/m^3$]

Table 1. Define variables table

1. Weather And Traffic Impact on Air Pollution

In this session, the model shows how the weather affects the traffic volume and how it affects air pollution.

DRIVERS OF TRAFFIC VOLUME

The horizontal bar chart below (Figure 1) displays the regression coefficients of the traffic model. Each bar in this chart indicates the marginal effect of a single variable on traffic volume while controlling all other variables.

- Precipitation shows that rainfall reduces traffic volume because it has the largest negative value on the bar chart.

- Day of week shows that traffic volume varies between days of the week. This indicates that weekdays have more traffic than weekends.

- Wind speed and snow depth are factors that also reduce traffic volume, possibly due to safety concerns.

- Month and hour show that traffic volume varies seasonally and depending on the time of day.

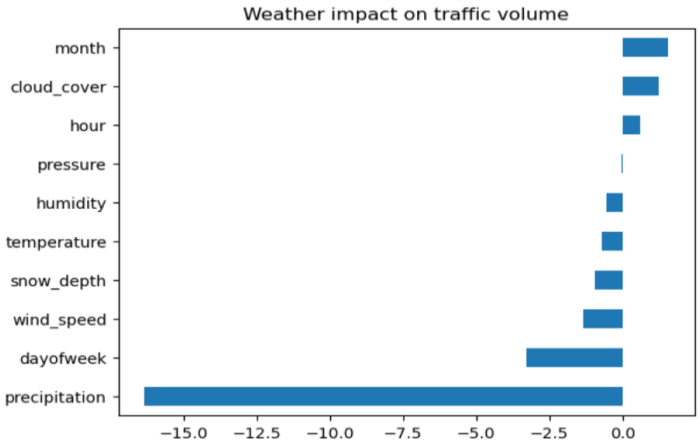


Figure 1. Weather effect on traffic volume

DRIVERS OF NO_2 AND BACK CARBON

The regression coefficients of the NO2 model (Figure 2) include weather variables and traffic volume: This indicates that wind_speed is responsible for dispersing emissions and is the element with the greatest negative impact on NO_2. Dayofweek shows low NO_2 concentrations at the end of the week. Temperature and cloud cover have a weak negative impact. Traffic volume, by itself, has a small negative impact. Precipitation has a positive impact, but clearly meteorological factors have the greatest impact on NO_2.

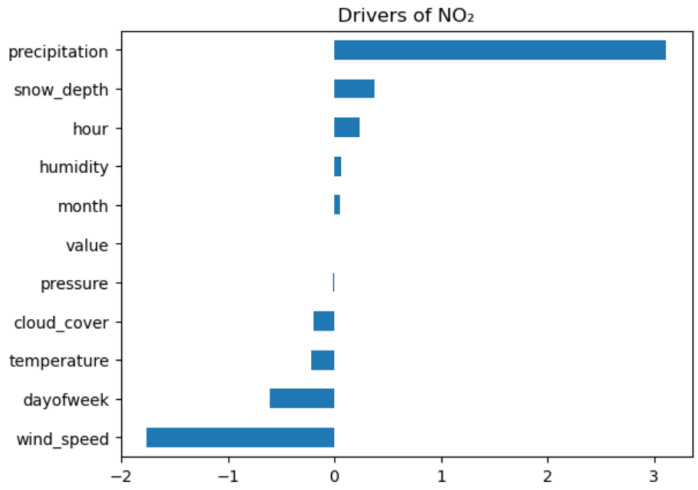


Figure 2. Drivers of NO_2

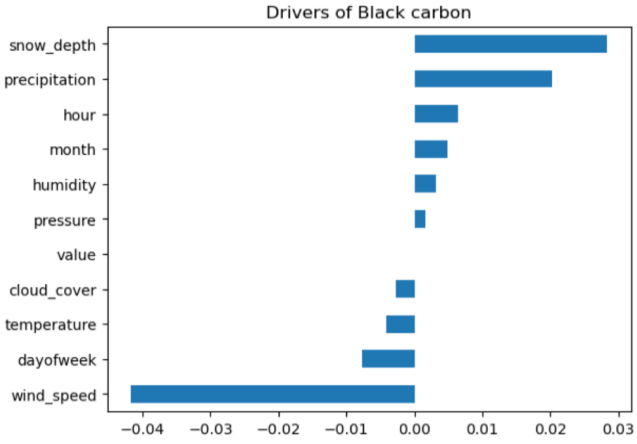


Figure 3. Drivers of Black Carbon

Black carbon (Figure 3) has a different driver compared to NO_2. Snow depth has the largest positive impact, followed by precipitation. The wind speed has a strong negative impact.

Together, Figures 2 and 3 illustrate that NO_2 and black carbon respond differently to meteorological conditions except for wind_speed.

2. Basic model analyzing techniques

Four different types of analysis were used for modelling: one-at-a-time sensitivity analysis, two-at-a-time sensitivity analysis, scenario analysis, and Monte Carlo.

ONE-AT-A-TIME SENSITIVITY ANALYSIS (SPIDER CHART)

All parameters in this model are adjusted for a 20% change, one parameter at a time. Figure 4 shows how NO2 changes when the input parameters are changed individually.

- Wind speed and pressure show the largest changes, with the steepest slopes, implying that how the atmosphere disperses pollutants is the main factor affecting NO2 concentration.

- Humidity and time-related parameters, such as hour and day of the week, also show significant effects, although much lower than wind speed and pressure.

- Changes in traffic volume show much lower effect than weather parameters.

The linear nature of the graphs also reflects the regression-based model, reinforcing the idea that weather is the main factor affecting NO2 concentration.

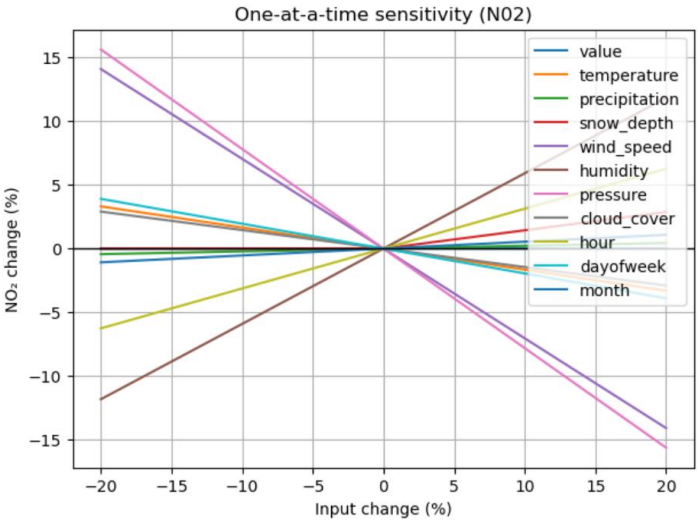


Figure 4. OAT sensitivity spider chart for NO_2

TWO-AT-A-TIME SENSITIVITY ANALYSIS (HEATMAP)

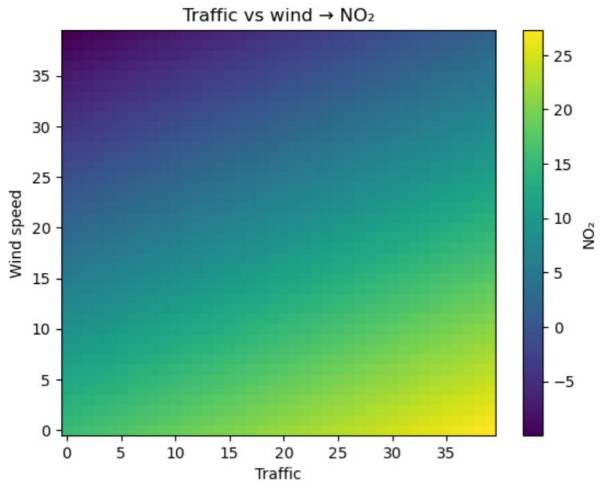


Figure 5. Heatmap

Based on the model, the wind_speed has been the most important factor in NO_2 pollution. In the heat map, I want to explore the effect of the traffic(value) and wind_speed in NO_2 pollution (Figure 5).

- NO_2 concentration increases with the traffic increases.

- NO_2 concentration decreases rapidly when wind speed increases.

- Pollution is present only with low wind speeds, even with moderate traffic volume.

SCENARIO ANALYSIS

Scenario analysis compares different levels of traffic reduction (R = 0.0, 0.2, 0.4) by comparing the median NO_2 concentration predicted by the model. The results indicate that a slight reduction in traffic reduces the median NO_2 concentration. However, the reduction was small, suggesting that the dominant factor in pollution variability is meteorological. The scenario analysis supports the fact that traffic reduction alone is not an effective factor in changing the level of pollution. (Table 2)

Traffic reduction (R)	NO_2 p5	NO_2 median	NO_2 p95
0.0	2.199	9.190	16.920
0.2	2.198	9.188	16.920
0.4	2.197	9.185	16.919

Table 2. Scenario analysis of traffic reduction

UNCERTAINTY ANALYSIS WITH MONTE CARLO SIMULATION

Uncertainty analysis was performed using Monte Carlo simulations. Results were obtained from 5000 hourly values in the dataset, under a traffic volume reduction policy R=0, and from pollution models simulating NO_2 and black carbon levels. The histogram shows the probability of different pollution level outcomes due to weather uncertainty.

The frequency histogram for NO_2 pollution levels shows a normal distribution curve with a long tail. The frequency histogram for black carbon levels shows a right skew. This suggests that periods of high pollution may occur due to weather uncertainty.

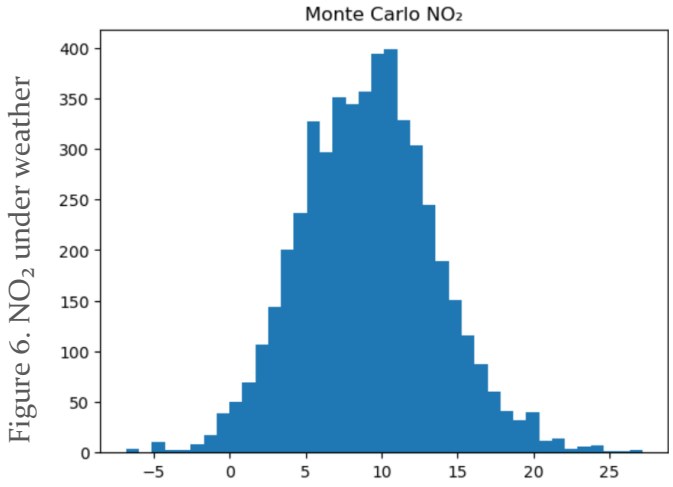


Figure 6. NO_2 under weather

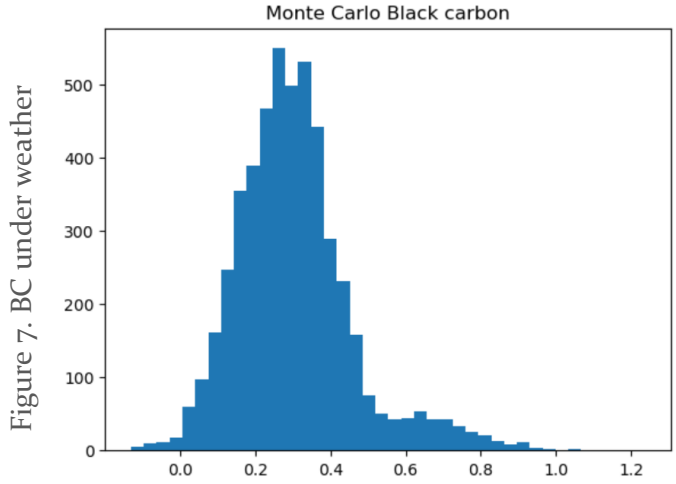


Figure 7. BC under weather

3. Modelling uncertainty and sampling

INPUT UNCERTAINTY

The histogram (Figure 8) shows the probability distribution for the uncertainty analysis of the major input variables. The traffic volume shows a triangular distribution resulting from the low, most likely, and high values of the dataset. The wind speed shows a normal distribution. The cloud cover shows a uniform distribution over the entire range. The precipitation shows an empirical distribution resulting from the dry periods and heavy rain.

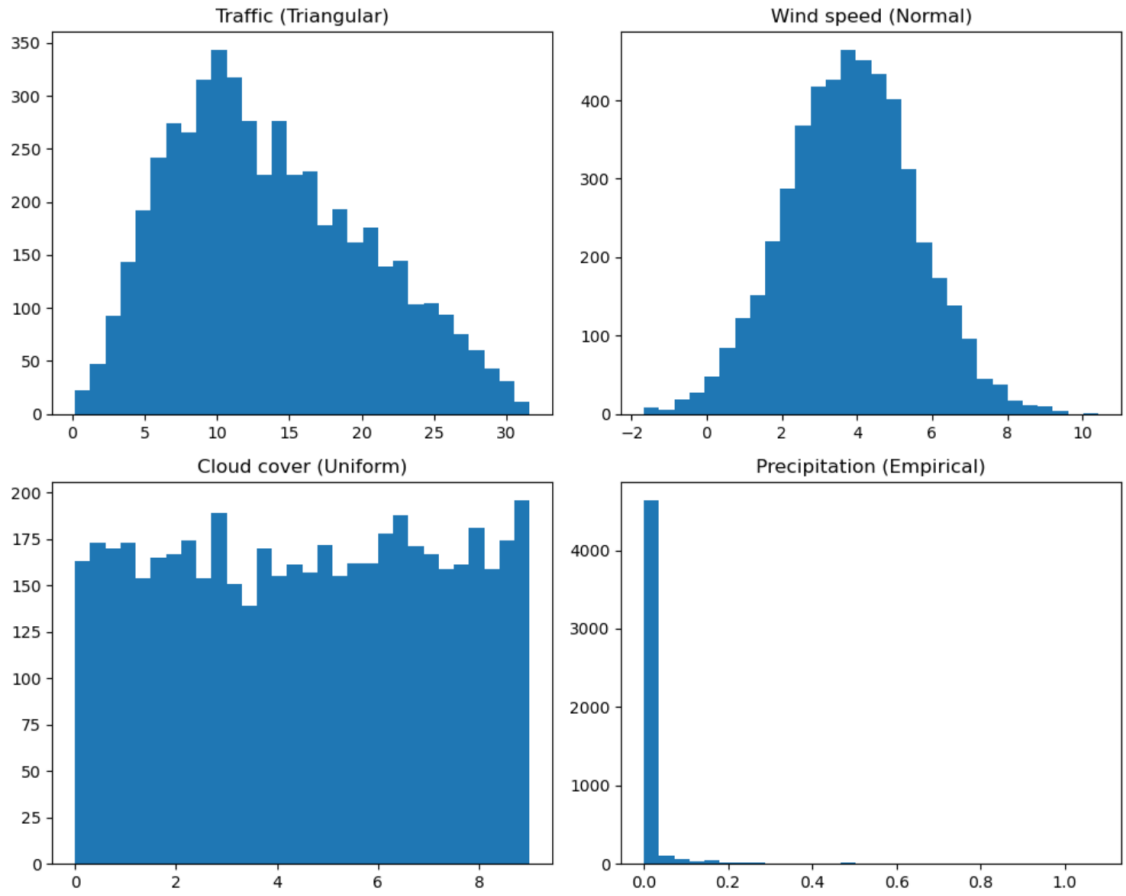


Figure 8. Histogram of uncertainty inputs

SAMPLING STRATEGIES

The scatter plots shown below (Figure 9) represent the sampling methods applied to traffic volume and wind speed. Simple random sampling is a method that can cause clustering and leave gaps in the input space, which can make uncertainty propagation less efficient. Latin hypercube sampling is a method that is better than simple random sampling, with samples that are more evenly distributed across the input space. Sobol quasi-Monte Carlo sampling

is the best, with the most comprehensive sampling, which is important for global sensitivity and uncertainty propagation analysis.

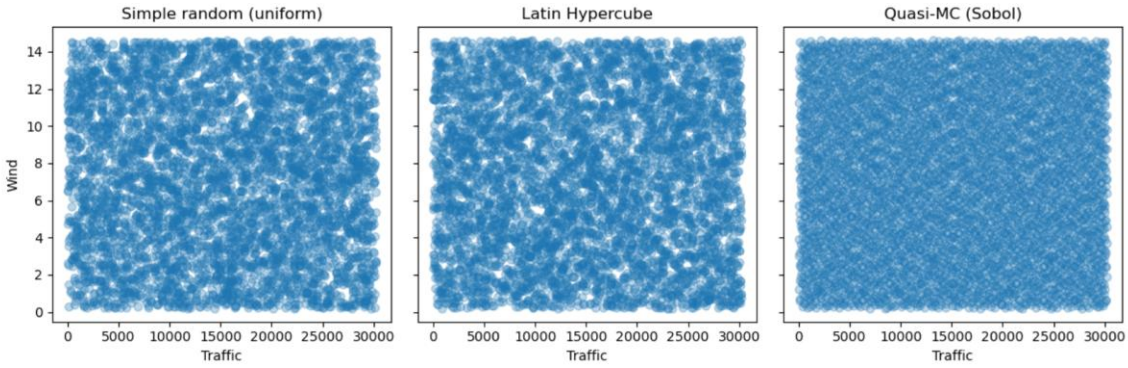


Figure 9. Scatter plots for sampling strategies

4. Global sensitivity analysis

Global sensitivity analysis was performed using Sobol's variance-based indices and the Morris screening method.

	Variable	mu_star
0	wind_speed	10.036
1	temperature	5.758
2	hour	5.068
3	snow_depth	4.103
4	dayofweek	3.609
5	humidity	2.923
6	cloud_cover	1.534
7	month	0.610
8	pressure	0.273
9	precipitation	0.218
10	value	0.013

Table 3. Morris

	Variable	S1	ST
3	wind_speed	0.495	0.493
0	temperature	0.171	0.158
7	hour	0.129	0.122
2	snow_depth	0.082	0.082
8	dayofweek	0.064	0.064
4	humidity	0.040	0.042
6	cloud_cover	0.009	0.012
9	month	0.002	0.002
5	pressure	0.001	0.000
1	precipitation	-0.000	0.000
10	value	0.000	0.000

Table 4. Sobol

Table 3 summarizes the Morris μ indices, which measure the level of importance for each input variable, based on the elementary effects calculated.

Table 4 summarizes the first order (S_1) and total-order (ST) Sobol indices, which were calculated using SALib.

Sobol and Morris analysis indicate wind speed as the most influential factor for the formation of NO_2, followed by temperature and temporal variables, with very low importance for traffic volume. The consistency between the two methods further supports the robustness of the sensitivity analysis.

5. Visualization

SIMDEC HISTOGRAM

The histogram created by the SimDec (Python) presents the predicted values for the NO_2 and Black Carbon concentrations for different combinations of weather and traffic conditions, categorized by the most influential variables and traffic volume. The colors indicate that the highest levels of pollution are generally associated with conditions, for example, when wind speed is low, and snow depth is high, which does not help with the dispersion of the pollution in the atmosphere.

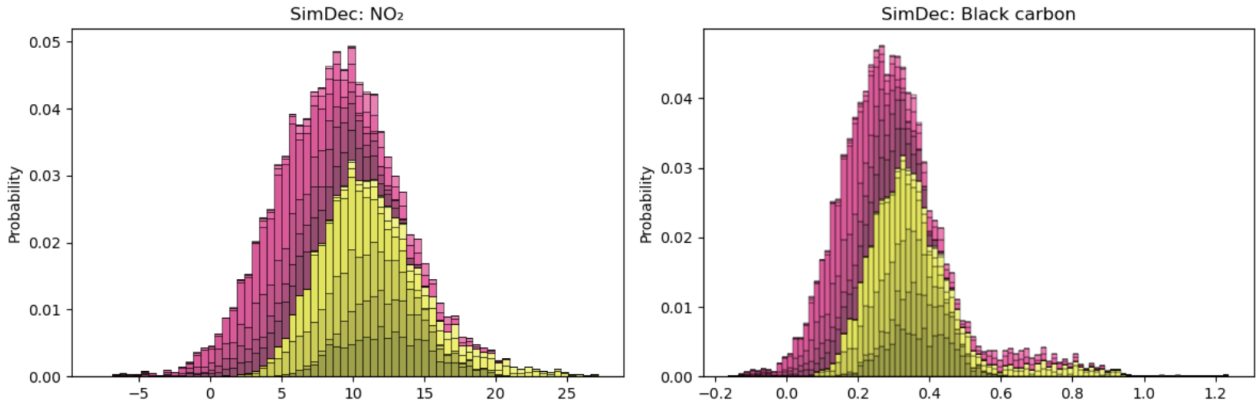


Figure 10. SimDec Histogram

The table in the legend indicates the average, range, and probability of each condition, making it clear that some of the highest levels of pollution are associated with rare conditions but are very significant when they do occur, and it is clear from the figure that the highest levels of air pollution are not associated with average conditions but with particular combinations of weather and traffic conditions.

wind_speed	snow_depth	temperature	traffic	N°	colour	std	min	mean	max	probability
low	low	low	low	16		3.10	4.54	12.31	21.06	0.10
			high	15		2.46	6.60	12.44	26.85	0.07
		high	low	14		2.98	0.87	9.66	19.98	0.12
			high	13		2.27	1.03	9.25	16.21	0.13
	high	low	low	12		4.28	7.10	15.88	26.73	0.04
			high	11		3.99	7.07	16.01	27.18	0.03
		high	low	10		2.20	6.98	11.27	13.56	0.00
			high	9		1.53	9.37	11.20	14.30	0.00
high	low	low	low	8		3.61	-7.92	7.03	16.12	0.09
			high	7		3.25	-5.05	7.00	14.93	0.07
		high	low	6		3.21	-6.56	5.28	12.89	0.09
			high	5		2.64	-6.19	4.68	12.30	0.15
	high	low	low	4		4.26	-4.31	10.77	20.79	0.05
			high	3		3.80	-1.37	11.49	26.35	0.04
		high	low	2		2.81	2.57	7.55	11.81	0.00
			high	1		4.10	-1.47	6.73	12.04	0.00

Table 5. Legend table

PARALLEL COORDINATES PLOT

The parallel coordinate plot (Figure 12) illustrates how traffic and weather interact to affect air pollution. Large values of NO_2 and Black Carbon are found mostly when wind speed is low, and snow depth is large, indicating that air is not being well dispersed. Large traffic levels do not cause large pollution levels unless accompanied by adverse weather conditions. When wind speed is higher, NO_2 and Black Carbon concentrations are low regardless of traffic levels. This indicates that the interaction of traffic and weather conditions is responsible for the high levels of air pollution, and thus, specific traffic management policies are more effective in low-wind winter conditions.

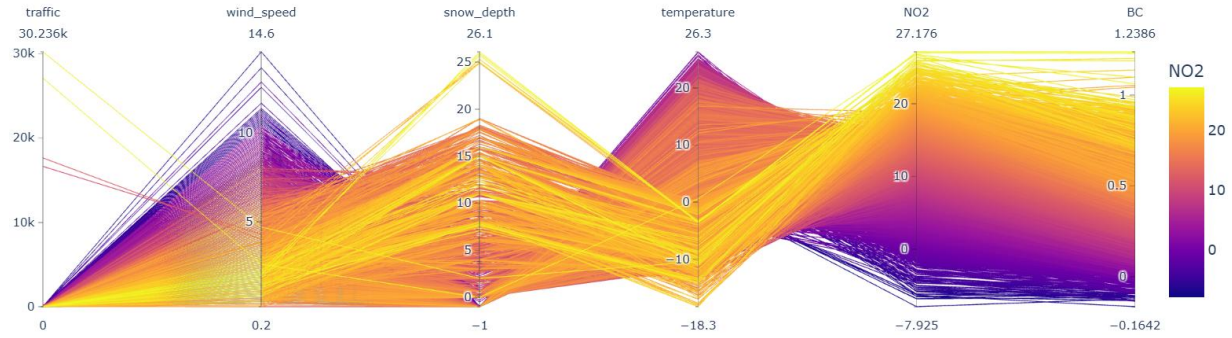


Figure 12. Parallel coordinate plot

6. Conclusions

The result of the analysis performed in this model shows that the most influential factors on air pollution are meteorological factors. The second most influential factor on air pollution is traffic volume. The wind speed is a major factor that influences the dispersion of pollutants. The low wind speed increases the concentration of NO_2. The traffic volume is a source of air pollution. The pollution level is high when traffic volume is combined with other unfavourable meteorological factors. The results of the scenario analysis show that reducing traffic volume uniformly does not have a major effect on the median concentration of NO_2.

The results of one-at-a-time and two-at-a-time sensitivity analyses show that wind speed is the most influential factor. The results of the Monte Carlo analysis show that the uncertainty of weather is the most influential factor on air pollution. The results of the global sensitivity analysis performed by the Sobol and Morris methods show that the results of the previous methods are correct. The most influential factor on the concentration of NO_2 is wind speed. The most influential factor on the concentration of black carbon is snow depth.

The results of the visualization methods, such as SimDec and parallel coordinates plots, show that the pollution level is high when measured in specific regions. The results of the

study show that the traffic management policies that take into account low wind speed and winter season are effective. The result also shows that the traffic management policies that consider a uniform reduction of traffic volume are not effective. The result also shows that the weather-aware policies are effective for air quality management.

References

Dataset:

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SimDec Visualization:

Kozlova, M. & Yeomans, J. S. (2025) 'Overview of SimDec applications', in Mariia Kozlova & Julian Scott Yeomans (eds) *Sensitivity Analysis for Business, Technology, and Policymaking*. 1st edition [Online]. Routledge. pp. 60–74.

Kozlova, M. (2025) *Sensitivity Analysis for Business, Technology, and Policymaking: Made Easy with Simulation Decomposition (SimDec)*. 1st edition. Mariia Kozlova & Julian Scott Yeomans (eds). [Online]. Oxford: Routledge.